

LAB EXPERIMENTS

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Set 1: Monthly Sales Data

Dataset: Monthly Sales

- 1. Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.**

Code :

```
# Load library
```

```
library(ggplot2)
```

```
# Dataset
```

```
sales <- data.frame(
```

```
  Month = factor(c("Jan", "Feb", "Mar", "Apr", "May", "Jun",  
                  "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"),
```

```
                  levels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun",  
                             "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")),
```

```
  Sales = c(12000, 15000, 18000, 17000, 22000, 25000,  
            24000, 27000, 30000, 32000, 35000, 40000)
```

```
)
```

```
# Line Chart
```

```
ggplot(sales, aes(x=Month, y=Sales, group=1)) +
```

```
  geom_line(color="blue", linewidth=1.2) +
```

```
  geom_point(color="red", size=3) +
```

```
  labs(
```

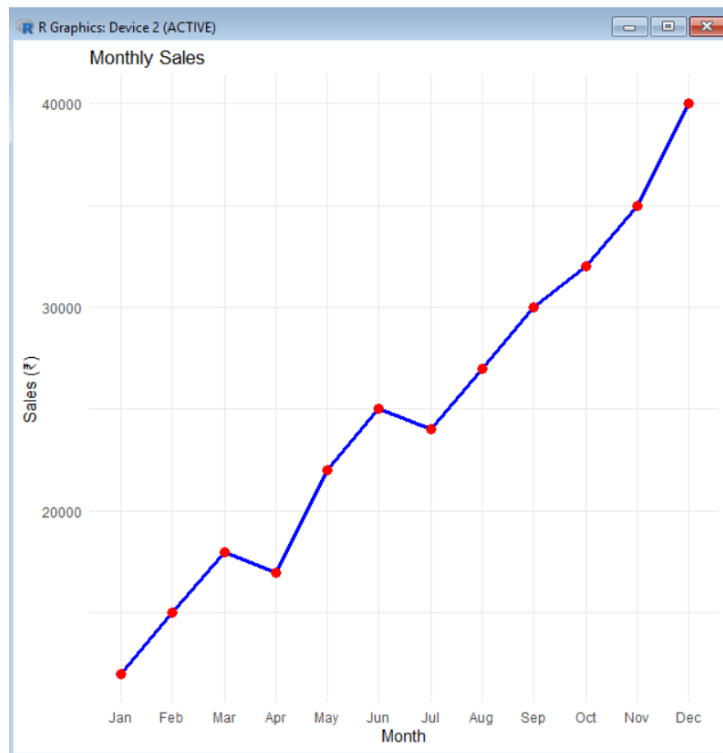
```
    title="Monthly Sales",
```

```

x="Month",
y="Sales (₹)"
) +
theme_minimal()

```

OUTPUT :



1.B .Generate a bar chart to display the top-selling products for the year. Label the chart elements.

CODE :

```

# Load library
library(ggplot2)

# Dataset
products <- data.frame(
  Product=c("Laptop","Mobile","Tablet","Headphones",
            "Smartwatch","Camera"),
  Sales=c(450,700,320,500,280,210)
)

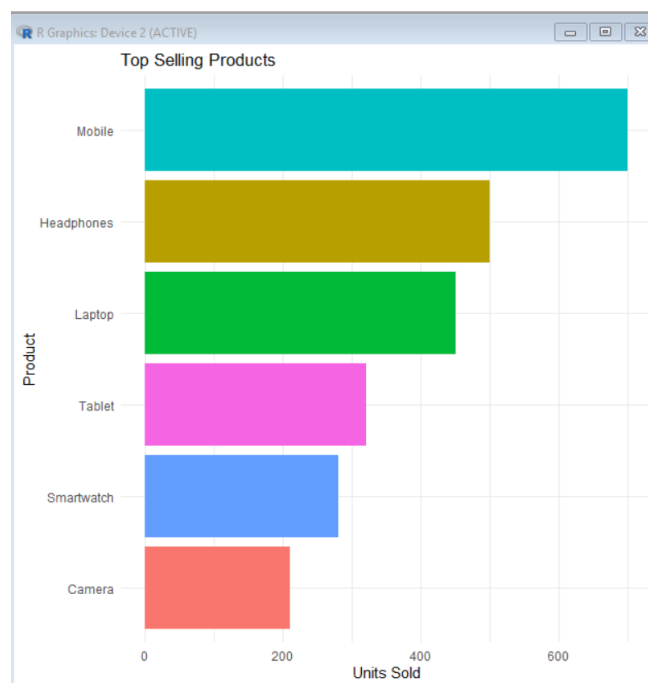
```

)

Bar Chart

```
ggplot(products, aes(x=reorder(Product, Sales), y=Sales, fill=Product)) +  
  geom_bar(stat="identity") +  
  coord_flip() +  
  labs(  
    title="Top Selling Products",  
    x="Product",  
    y="Units Sold"  
  ) +  
  theme_minimal() +  
  theme(legend.position="none")
```

OUTPUT :



1.C. Develop a scatter plot to explore the relationship between advertising budget and monthly sales.

Explain the insights drawn from the scatter plot.

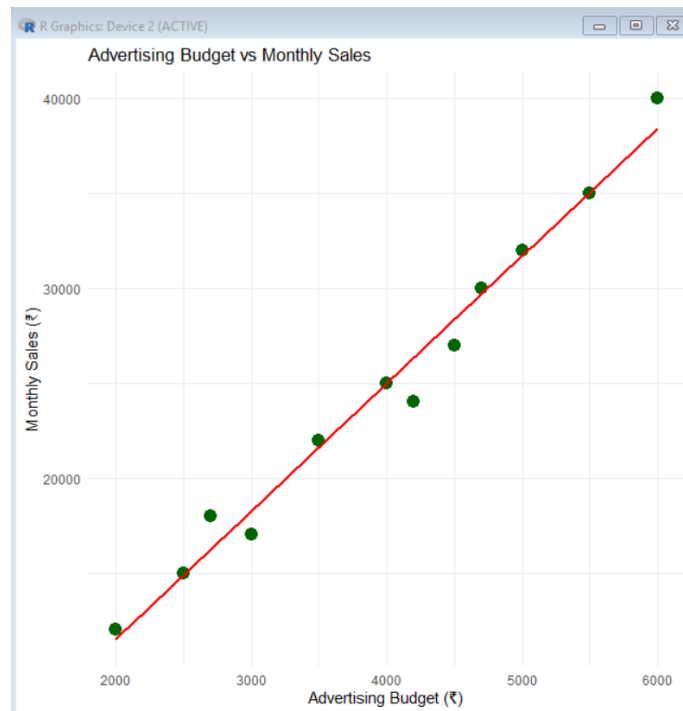
CODE :

```
# Load library
library(ggplot2)

# Dataset
advertising <- data.frame(
  Month=c("Jan","Feb","Mar","Apr","May","Jun",
          "Jul","Aug","Sep","Oct","Nov","Dec"),
  Budget=c(2000,2500,2700,3000,3500,4000,
           4200,4500,4700,5000,5500,6000),
  Sales=c(12000,15000,18000,17000,22000,25000,
          24000,27000,30000,32000,35000,40000)
)

# Scatter Plot
ggplot(advertising, aes(x=Budget, y=Sales)) +
  geom_point(color="darkgreen", size=4) +
  geom_smooth(method="lm", se=FALSE, color="red") +
  labs(
    title="Advertising Budget vs Monthly Sales",
    x="Advertising Budget (₹)",
    y="Monthly Sales (₹)"
  ) +
  theme_minimal()
```

OUTPUT :



Insights from Scatter Plot

- There is a **positive relationship** between advertising budget and monthly sales.
- As the advertising budget increases, sales also tend to increase.
- The trend line shows a **strong positive correlation**.
- This suggests that increased advertising may contribute to higher sales.

Program 4: Interactive Dashboard (Line Chart + Bar Chart)

CODE :

```
# Install packages if needed
# install.packages("flexdashboard")
# install.packages("plotly")

library(flexdashboard)
library(plotly)
library(ggplot2)

# Dataset 1
```

```

sales <- data.frame(
  Month=factor(c("Jan","Feb","Mar","Apr","May","Jun",
    "Jul","Aug","Sep","Oct","Nov","Dec"),
    levels=c("Jan","Feb","Mar","Apr","May","Jun",
    "Jul","Aug","Sep","Oct","Nov","Dec")),
  Sales=c(12000,15000,18000,17000,22000,25000,
    24000,27000,30000,32000,35000,40000)
)

```

Dataset 2

```

products <- data.frame(
  Product=c("Laptop","Mobile","Tablet","Headphones",
    "Smartwatch","Camera"),
  Sales=c(450,700,320,500,280,210)
)

```

Line Chart

```

line_plot <- ggplot(sales, aes(Month, Sales, group=1)) +
  geom_line(color="blue", linewidth=1.2) +
  geom_point(size=3, color="red") +
  labs(title="Monthly Sales")

```

Bar Chart

```

bar_plot <- ggplot(products,
  aes(reorder(Product, Sales), Sales, fill=Product)) +
  geom_bar(stat="identity") +
  coord_flip() +
  labs(title="Top Selling Products") +

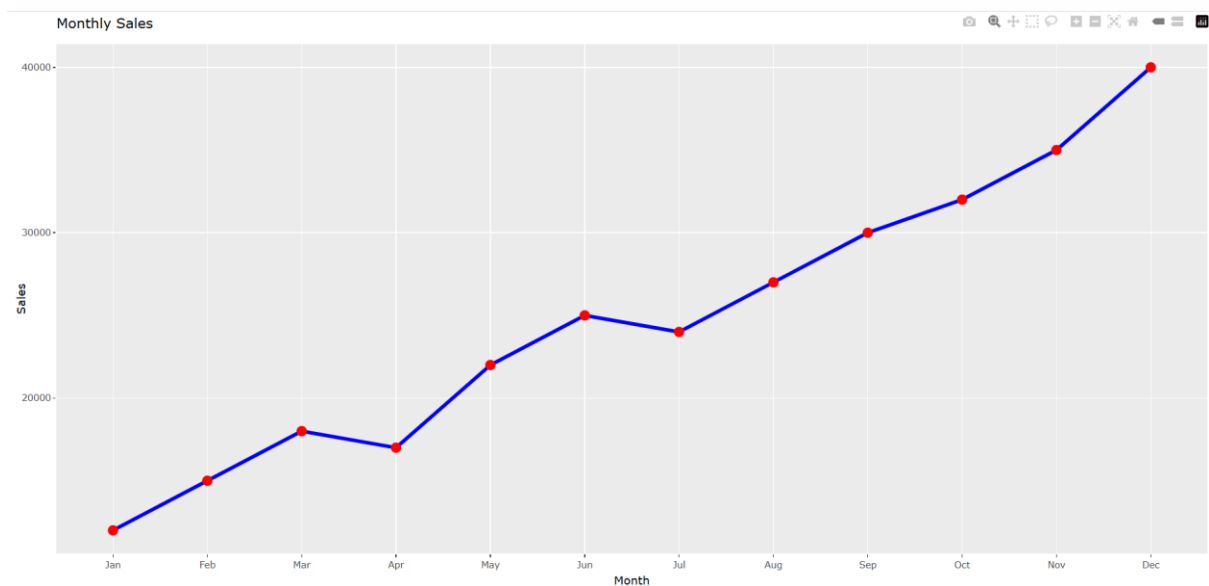
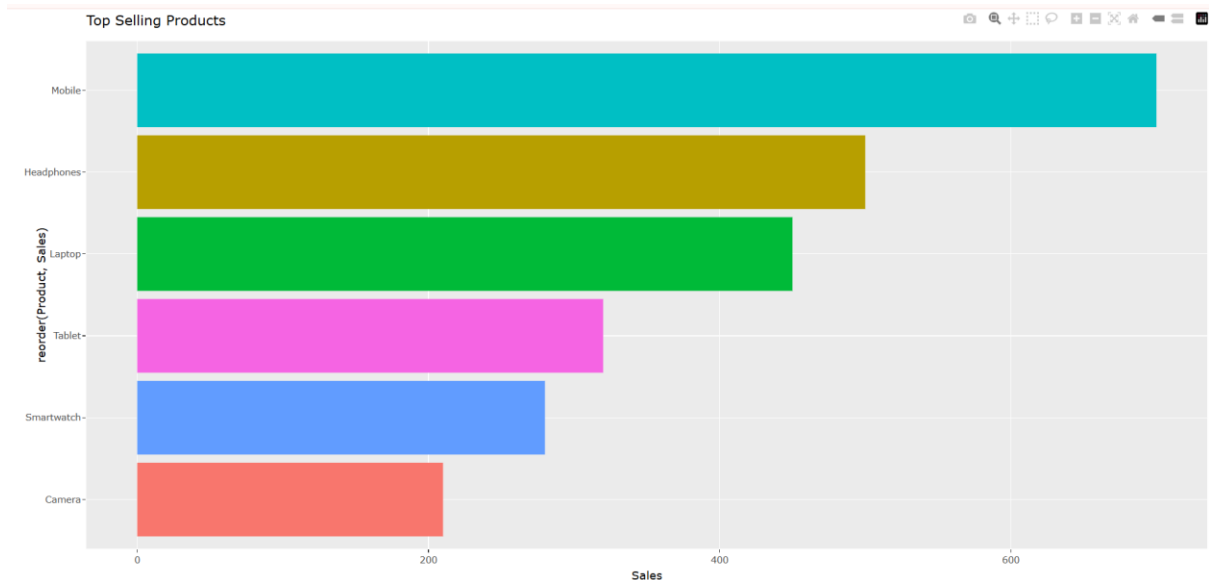
```

```
theme(legend.position="none")
```

```
ggplotly(line_plot)
```

```
ggplotly(bar_plot)
```

OUTPUT :



SET 2.Program 1: Histogram – Customer Age Distribution

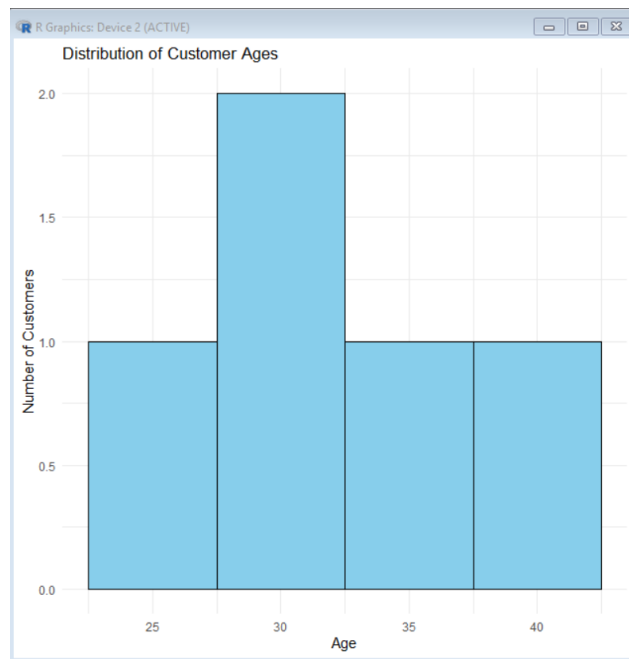
CODE :

```
library(ggplot2)
```

```
customer <- data.frame(
  CustomerID = c(1,2,3,4,5),
  Age = c(25,30,35,28,40),
  Satisfaction = c(4,5,3,4,5)
)

ggplot(customer, aes(x = Age)) +
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +
  labs(
    title = "Distribution of Customer Ages",
    x = "Age",
    y = "Number of Customers"
  ) +
  theme_minimal()
```

OUTPUT :



2.B : Pie Chart – Customer Satisfaction Distribution

CODE :

```
library(ggplot2)
```



```
# Dataset
```

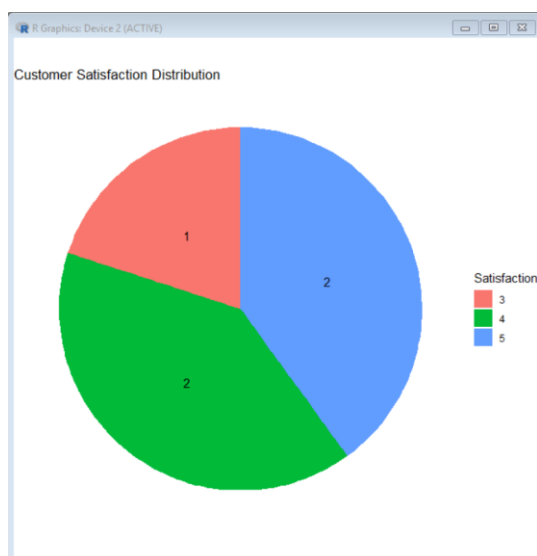
```
customer <- data.frame(  
  CustomerID = c(1,2,3,4,5),  
  Age = c(25,30,35,28,40),  
  Satisfaction = c(4,5,3,4,5)  
)
```

```
# Count satisfaction scores
```

```
pie_data <- as.data.frame(table(customer$Satisfaction))  
colnames(pie_data) <- c("Satisfaction", "Count")
```

```
# Pie Chart
```

```
ggplot(pie_data, aes(x = "", y = Count, fill = Satisfaction)) +  
  geom_bar(stat = "identity", width = 1) +  
  coord_polar("y") +  
  geom_text(aes(label = Count),  
            position = position_stack(vjust = 0.5)) +  
  labs(title = "Customer Satisfaction Distribution") +  
  theme_void()
```



2.C: Stacked Bar Chart – Satisfaction by Age Group

CODE :

```
library(ggplot2)

# Dataset

customer <- data.frame(

  CustomerID = c(1,2,3,4,5),

  Age = c(25,30,35,28,40),

  Satisfaction = c(4,5,3,4,5)

)

# Create Age Groups

customer$AgeGroup <- cut(

  customer$Age,

  breaks = c(20,30,40,50),

  labels = c("21-30","31-40","41-50"),

  include.lowest = TRUE

)

# Stacked Bar Chart

ggplot(customer,

  aes(x = AgeGroup,

      fill = factor(Satisfaction))) +

geom_bar() +

labs(

  title = "Customer Satisfaction by Age Group",

  x = "Age Group",

  y = "Number of Customers",
```

```
fill = "Satisfaction"
```

```
) +
```

```
theme_minimal()
```



2.D .Word Cloud – Customer Feedback

CODE: library(wordcloud)

library(tm)

Customer Feedback

```
feedback <- c(
```

```
"Excellent service and friendly staff",
```

```
"Very satisfied with product quality",
```

```
"Good experience and quick delivery",
```

```
"Friendly support and excellent response",
```

```
"Satisfied with service and quality"
```

```
)
```

```
docs <- Corpus(VectorSource(feedback))
```

```
# Convert to lowercase
docs <- tm_map(docs, content_transformer(tolower))

# Remove punctuation
docs <- tm_map(docs, removePunctuation)

# Remove numbers
docs <- tm_map(docs, removeNumbers)

# Remove common words
docs <- tm_map(docs, removeWords, stopwords("english"))

# Create Term Matrix
dtm <- TermDocumentMatrix(docs)

# Convert to Matrix
m <- as.matrix(dtm)

# Word Frequency
word_freq <- sort(rowSums(m), decreasing = TRUE)

# Word Cloud
wordcloud(
  words = names(word_freq),
  freq = word_freq,
  min.freq = 1,
  random.order = FALSE,
  colors = rainbow(length(word_freq))
)
```



SET 3 : Employee Performance Evaluation

1. Line Chart – Employee Performance Trend

```
library(ggplot2)
```

```
employee <- data.frame(
```

```
  EmployeeID = c(1,2,3,4,5),
```

```
  Department = c("Sales","HR","Marketing","Sales","HR"),
```

```
  YearsService = c(5,3,7,4,2),
```

```
  Performance = c(85,92,78,90,76)
```

```
)
```

```
ggplot(employee, aes(x=YearsService, y=Performance, color=Department)) +
```

```
  geom_line(linewidth=1) +
```

```
  geom_point(size=3) +
```

```
  labs(
```

```
    title="Employee Performance Trend",
```

```
    x="Years of Service",
```

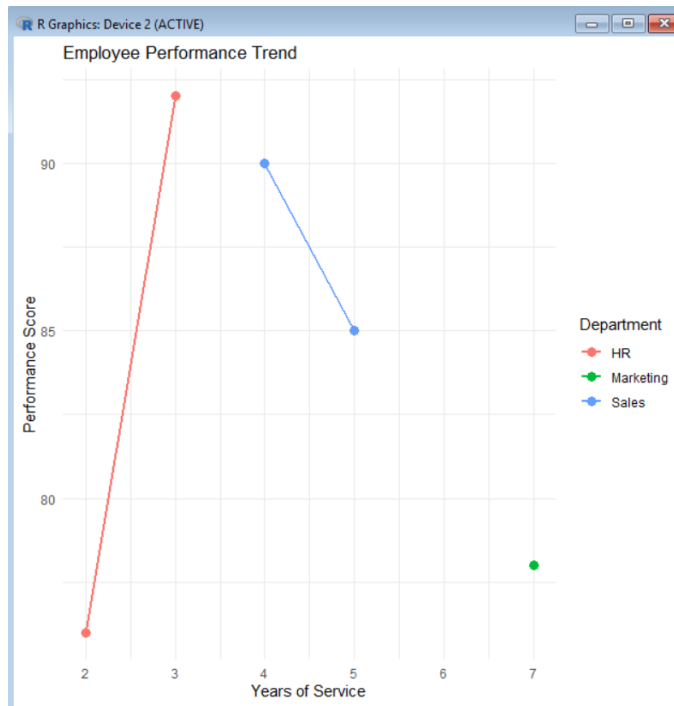
```
    y="Performance Score",
```

```
    color="Department"
```

```
) +
```

```
theme_minimal()
```

OUTPUT :



2. Bar Chart – Employees by Department

```
library(ggplot2)
```

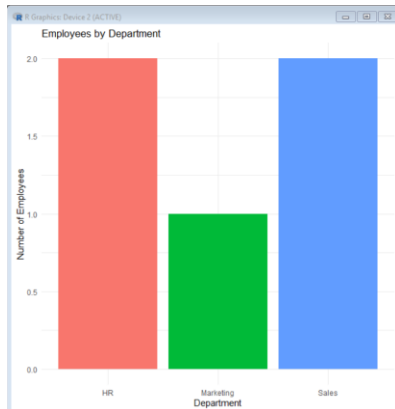
```
employee <- data.frame(  
  EmployeeID=c(1,2,3,4,5),  
  Department=c("Sales", "HR", "Marketing", "Sales", "HR")  
)
```

```
ggplot(employee, aes(x=Department, fill=Department)) +  
  geom_bar() +  
  labs(  
    title="Employees by Department",  
    x="Department",  
    y="Number of Employees"
```

```
) +
```

```
theme_minimal() +
```

```
theme(legend.position="none")
```



3. Scatter Plot – Years of Service vs Performance

```
library(ggplot2)
```

```
employee <- data.frame(
```

```
  YearsService=c(5,3,7,4,2),
```

```
  Performance=c(85,92,78,90,76)
```

```
)
```

```
ggplot(employee, aes(YearsService, Performance)) +
```

```
  geom_point(size=4, color="blue") +
```

```
  geom_smooth(method="lm", se=FALSE, color="red") +
```

```
  labs(
```

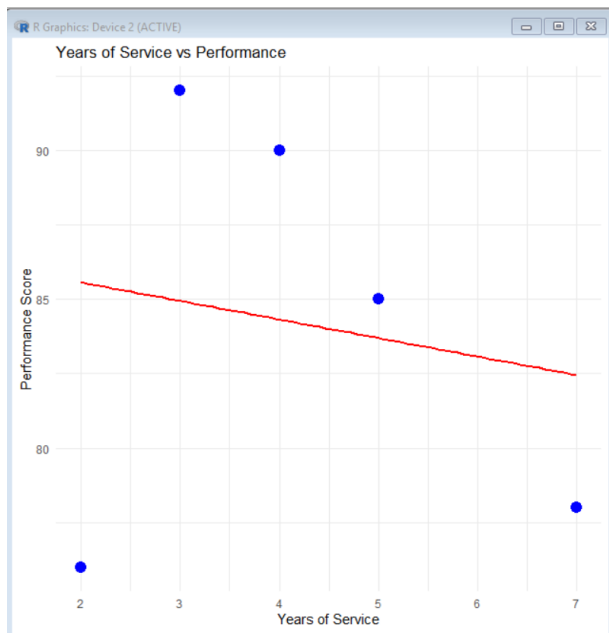
```
    title="Years of Service vs Performance",
```

```
    x="Years of Service",
```

```
    y="Performance Score"
```

```
) +
```

```
theme_minimal()
```



4. Employee Performance Dashboard (R Shiny)

```
library(shiny)
```

```
library(ggplot2)
```

```
employee <- data.frame(
```

```
  EmployeeID = c(1,2,3,4,5),
```

```
  Department = c("Sales","HR","Marketing","Sales","HR"),
```

```
  YearsService = c(5,3,7,4,2),
```

```
  Performance = c(85,92,78,90,76)
```

```
)
```

```
ui <- fluidPage(
```

```
  titlePanel("Employee Performance Dashboard"),
```

```
  sidebarLayout(
```

```
    sidebarPanel(
```

```
      selectInput("dept", "Select Department",
```

```
        choices=c("All",unique(employee$Department)))
```

```
    ),
```

```
    mainPanel(
```

```
      plotOutput("linePlot"),
```



```

    plotOutput("barPlot")
  )
)
)

server <- function(input, output){

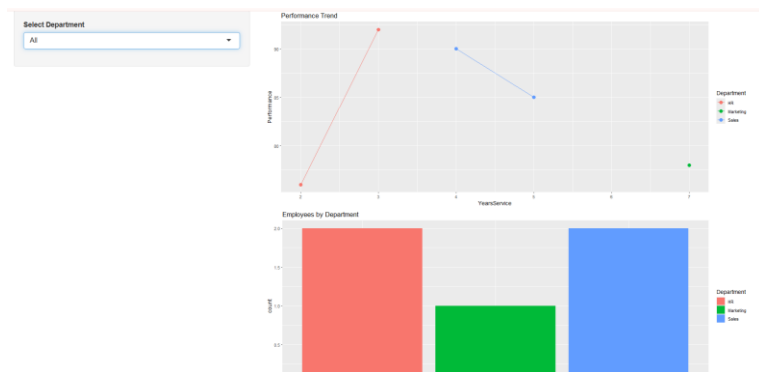
  data <- reactive({
    if(input$dept=="All")
      employee
    else
      subset(employee, Department==input$dept)
  })

  output$linePlot <- renderPlot({
    ggplot(data(), aes(YearsService, Performance,
                      color=Department, group=Department))+
      geom_line()+
      geom_point(size=3)+
      labs(title="Performance Trend")
  })

  output$barPlot <- renderPlot({
    ggplot(data(), aes(Department, fill=Department))+
      geom_bar()+
      labs(title="Employees by Department")
  })
}

```

```
shinyApp(ui, server)
```



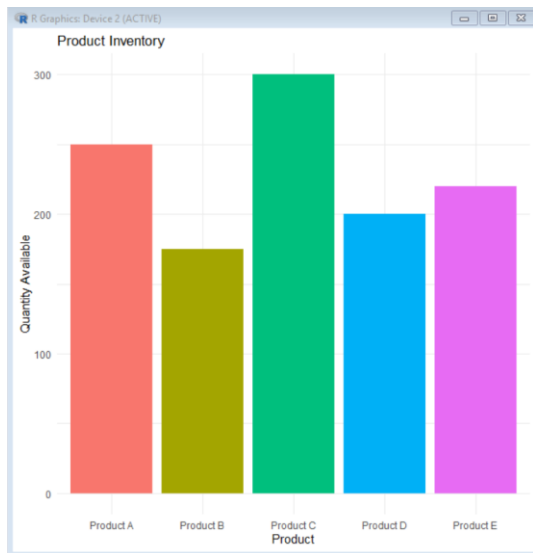
SET 4: Product Inventory Management

1. Bar Chart – Product Quantity

```
library(ggplot2)
```

```
inventory <- data.frame(
  Product=c("Product A","Product B","Product C","Product D","Product E"),
  Quantity=c(250,175,300,200,220)
)
```

```
ggplot(inventory,
  aes(Product, Quantity, fill=Product)) +
  geom_bar(stat="identity") +
  labs(
    title="Product Inventory",
    x="Product",
    y="Quantity Available"
  ) +
  theme_minimal() +
  theme(legend.position="none")
```

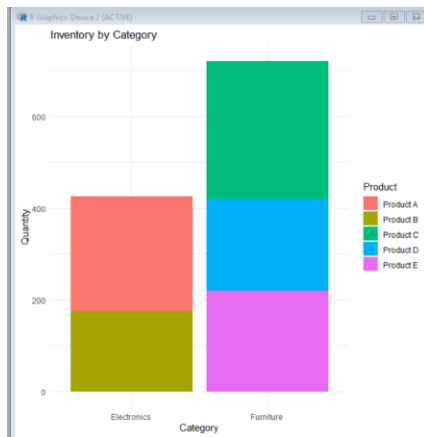


2. Stacked Bar Chart – Quantity by Category

```
library(ggplot2)
```

```
inventory <- data.frame(
  Product=c("Product A","Product B","Product C","Product D","Product E"),
  Category=c("Electronics","Electronics",
             "Furniture","Furniture","Furniture"),
  Quantity=c(250,175,300,200,220)
)
```

```
ggplot(inventory,
  aes(Category, Quantity, fill=Product)) +
  geom_bar(stat="identity") +
  labs(
    title="Inventory by Category",
    x="Category",
    y="Quantity"
  ) +
  theme_minimal()
```



3. Scatter Plot – Price vs Quantity

```
library(ggplot2)
```

```
inventory <- data.frame(
```

```
  Product=c("Product A","Product B","Product C","Product D","Product E"),
```

```
  Price=c(1200,900,1500,1000,1100),
```

```
  Quantity=c(250,175,300,200,220)
```

```
)
```

```
ggplot(inventory,
```

```
  aes(Price, Quantity)) +
```

```
geom_point(size=4, color="darkgreen") +
```

```
geom_smooth(method="lm", se=FALSE, color="red") +
```

```
labs(
```

```
  title="Price vs Quantity",
```

```
  x="Price",
```

```
  y="Quantity"
```

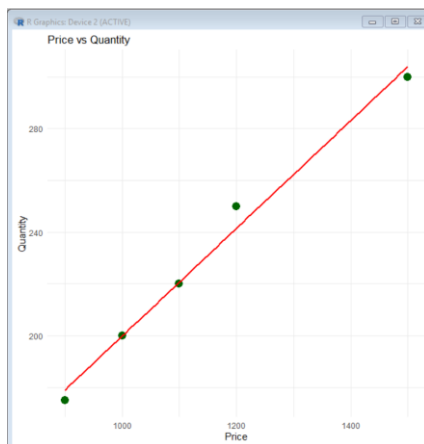
```
) +
```

```
theme_minimal()
```

Finding

- No strong relationship exists between price and available quantity.

- Since the dataset is small, no definite trend can be concluded.



Set 4 – Product Inventory Dashboard (R Shiny)

```
library(shiny)
```

```
library(ggplot2)
```

```
inventory <- data.frame(
  Product=c("Product A","Product B","Product C","Product D","Product E"),
  Category=c("Electronics","Electronics",
             "Furniture","Furniture","Furniture"),
  Quantity=c(250,175,300,200,220)
)
```

```
ui <- fluidPage(
```

```
  titlePanel("Inventory Dashboard"),
```

```
  sidebarLayout(
```

```
    sidebarPanel(
```

```
      selectInput("cat","Category",
```

```
      choices=c("All",unique(inventory$Category)))
```

```
),
```

```
mainPanel(  
  plotOutput("bar"),  
  plotOutput("stack")  
)
```

```
)
```

```
)
```

```
)
```

```
server <- function(input, output){
```

```
  data <- reactive({
```

```
    if(input$cat=="All")
```

```
      inventory
```

```
    else
```

```
      subset(inventory, Category==input$cat)
```

```
  })
```

```
  output$bar <- renderPlot({
```

```
    ggplot(data(),aes(Product,Quantity,fill=Product))+
```

```
    geom_bar(stat="identity")+
```

```
    labs(title="Product Quantity")
```

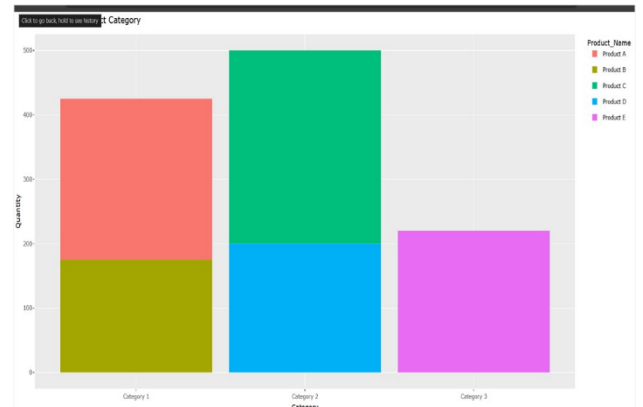
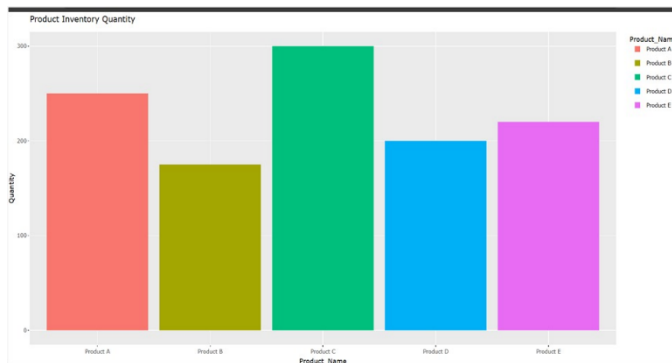
```
  })
```

```
  output$stack <- renderPlot({
```

```

ggplot(data(),aes(Category,Quantity,fill=Product))+
geom_bar(stat="identity")+
labs(title="Inventory by Category")
})
}
shinyApp(ui,server)

```



SET 5: Website Analytics

1. Line Chart – Daily Page Views

```
library(ggplot2)
```

```

traffic <- data.frame(
  Date=as.Date(c("2023-01-01","2023-01-02","2023-01-03",
    "2023-01-04","2023-01-05")),
  PageViews=c(1500,1600,1400,1650,1800),
  CTR=c(2.3,2.7,2.0,2.4,2.6)
)

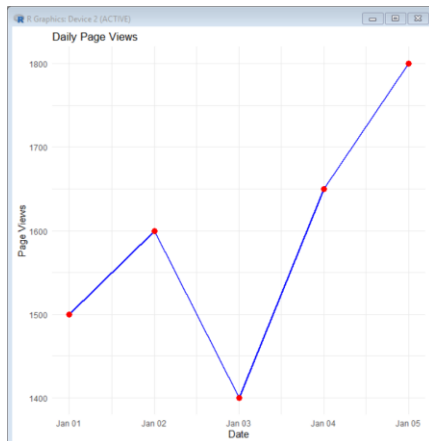
```

```

ggplot(traffic,
  aes(Date, PageViews, group=1)) +
  geom_line(color="blue", linewidth=1) +
  geom_point(size=3, color="red") +

```

```
labs(
  title="Daily Page Views",
  x="Date",
  y="Page Views"
) +
theme_minimal()
```



2. Bar Chart – Click Through Rate

```
library(ggplot2)

traffic <- data.frame(
  Date=as.Date(c("2023-01-01","2023-01-02","2023-01-03",
    "2023-01-04","2023-01-05")),
  CTR=c(2.3,2.7,2.0,2.4,2.6)
)

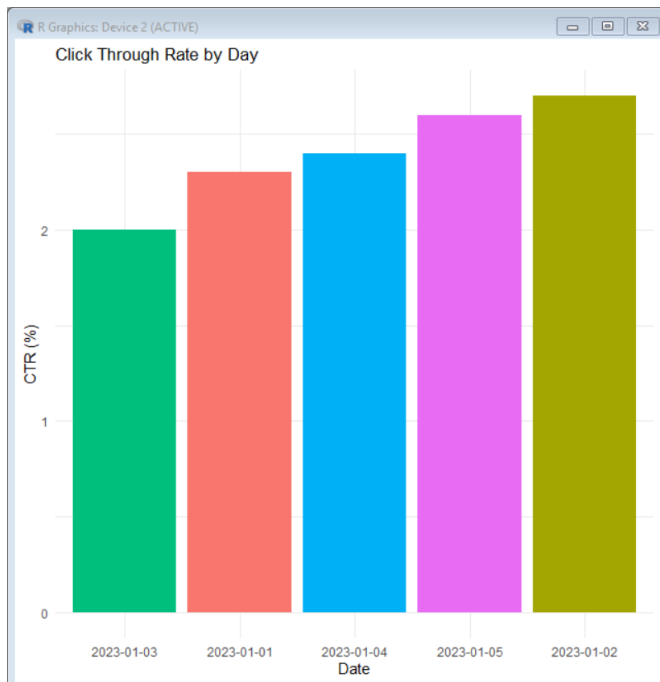
ggplot(traffic,
  aes(reorder(Date, CTR), CTR,
    fill=factor(Date))) +
  geom_bar(stat="identity") +
  labs(
    title="Click Through Rate by Day",
```



```

x="Date",
y="CTR (%)"
) +
theme_minimal() +
theme(legend.position="none")

```



3. Stacked Area Chart – User Interactions

```
library(ggplot2)
```

```
library(tidyr)
```

```

interaction <- data.frame(
  Date=as.Date(c("2023-01-01","2023-01-02","2023-01-03",
    "2023-01-04","2023-01-05")),
  Likes=c(120,150,130,170,180),
  Shares=c(40,50,35,60,70),
  Comments=c(25,30,20,35,40)
)

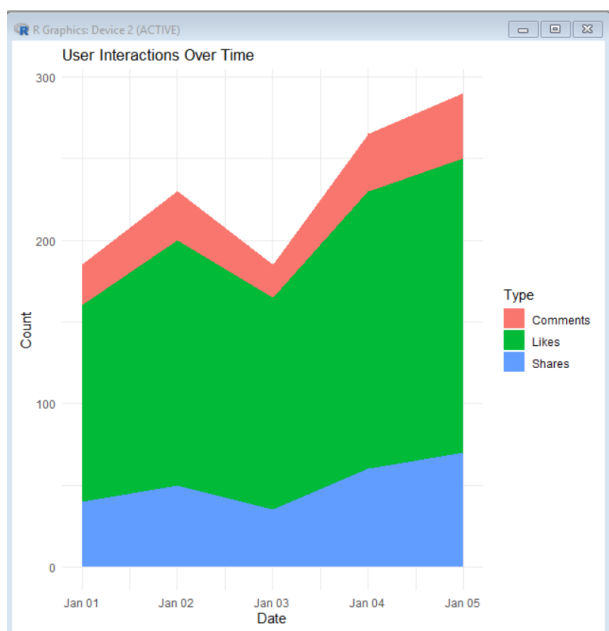
```

```

interaction_long <- pivot_longer(
  interaction,
  cols=c(Likes,Shares,Comments),
  names_to="Type",
  values_to="Count"
)

ggplot(interaction_long,
  aes(Date, Count, fill=Type)) +
  geom_area() +
  labs(
    title="User Interactions Over Time",
    x="Date",
    y="Count"
  ) +
  theme_minimal()

```



5.D.Website Analytics Dashboard (R Shiny)

```

library(shiny)

library(ggplot2)

```

```
traffic <- data.frame(  
  Date=as.Date(c("2023-01-01","2023-01-02","2023-01-03",  
    "2023-01-04","2023-01-05")),  
  PageViews=c(1500,1600,1400,1650,1800),  
  CTR=c(2.3,2.7,2.0,2.4,2.6)  
)
```

```
ui <- fluidPage(  
  
  titlePanel("Website Analytics Dashboard"),  
  
  mainPanel(  
    plotOutput("line"),  
    plotOutput("bar")  
  )  
  
)
```

```
server <- function(input, output){  
  
  output$line <- renderPlot({  
  
    ggplot(traffic,aes(Date,PageViews))+  
    geom_line(color="blue")+  
    geom_point(size=3,color="red")+  
    labs(title="Daily Page Views")
```

```
})
```

```
output$bar <- renderPlot({
```

```
  ggplot(traffic,aes(as.factor(Date),CTR,fill=as.factor(Date)))+
```

```
  geom_bar(stat="identity")+
```

```
  labs(title="Click Through Rate",
```

```
        x="Date",
```

```
        y="CTR (%)")+
```

```
  theme(legend.position="none")
```

```
})
```

```
}
```

```
shinyApp(ui,server)
```

